



OMNİKON

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01 // 06

TEAM NAME

RETINASIGHT AI

MEMBERS

Soham Joshi

Chitrangad Sapate

THEME

BIO / HEALTHCARE / AI · OMNI_BIOTECH_1

PROBLEM STATEMENT

Early Detection of Diabetic Retinopathy

Enable early, low-cost diabetic retinopathy screening and referral in rural areas using smartphone-based retinal image analysis.

SMARTPHONE IMAGING

AI-ASSISTED SCREENING

EXPLAINABLE AI

OFFLINE-FIRST

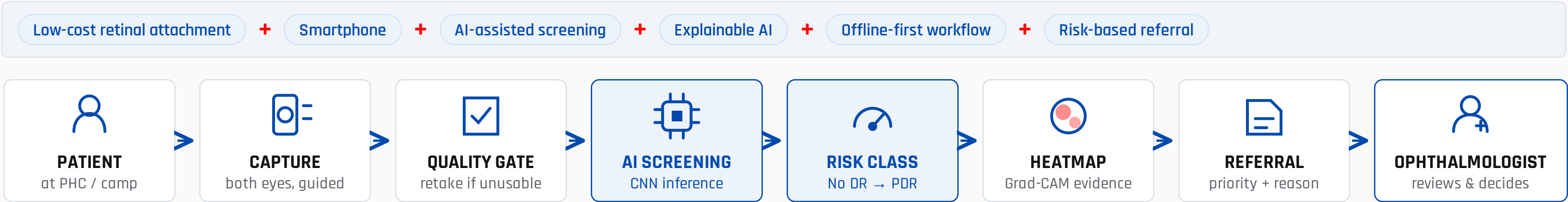
RISK-BASED REFERRAL

Screen early. Refer early. Prevent avoidable blindness.

Proposed Solution

Smartphone-based AI-assisted retinal screening & referral platform for rural, low-resource care

SCREENING & CLINICAL DECISION SUPPORT
Not an autonomous diagnostic system



HOW IT WORKS

- 1

Capture. Low-cost smartphone retinal attachment captures left/right eye images with on-screen alignment guidance.
- 2

Quality gate. Blur, lighting, framing and retinal visibility are checked before inference; unusable images trigger a retake prompt.
- 3

AI screening. A transfer-learned CNN estimates DR severity across five screening categories with a confidence score.
- 4

Explain. Grad-CAM highlights the retinal regions that influenced the prediction, so the output is inspectable.
- 5

Refer. Risk band maps to a referral priority – routine monitoring, consultation, or urgent ophthalmology referral.

HOW IT SOLVES THE PROBLEM

- LOW COST**

Smartphone-based imaging reduces dependence on expensive dedicated fundus equipment.
- RURAL READY**

Offline-first workflow keeps screening running where connectivity is unreliable or absent.
- SCREEN → REFER**

Connects screening directly to specialist referral, review and follow-up instead of ending at a result.

WHY IT IS UNIQUE

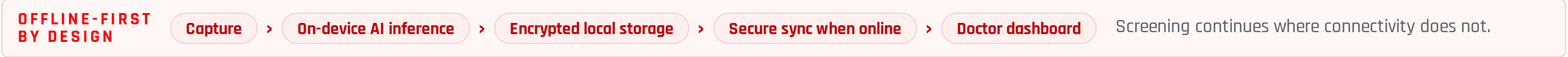
- ▶ **Offline-first AI**

▶ **Automated image-quality gate**

▶ **Explainable AI (Grad-CAM)**

▶ **Risk-based referral engine**

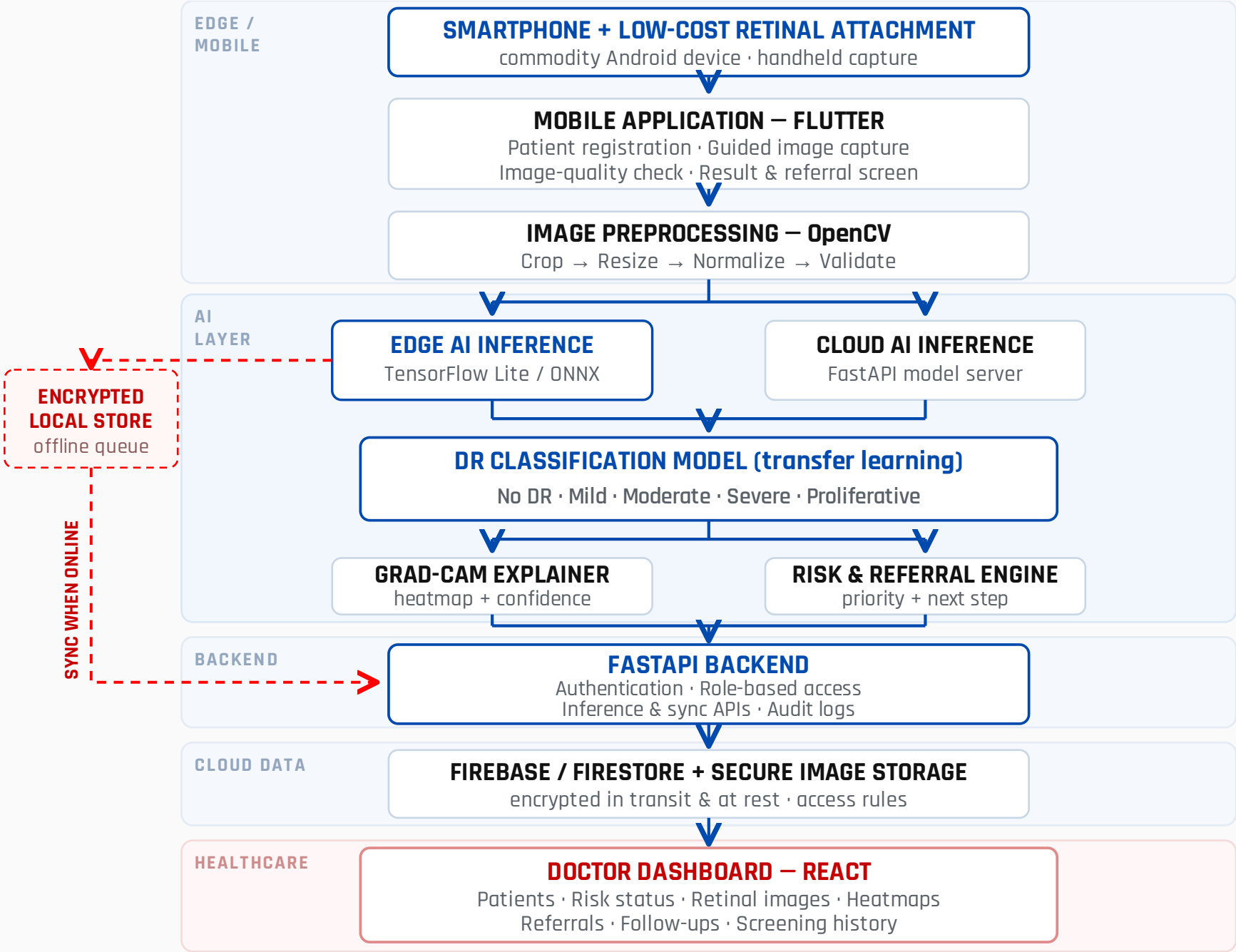
▶ **Low-cost smartphone imaging**
- The innovation is the workflow, not the model alone. Quality gating, edge inference, explainability and referral routing are what make screening usable at a PHC.



End-to-End System Architecture

Edge / mobile → AI layer → backend → cloud data → healthcare layer, with an offline synchronisation path

Primary flow Offline / deferred sync



TECHNOLOGY STACK	
MOBILE	Flutter (Dart) – Android / iOS
AI / ML	Python · PyTorch / TensorFlow
MODELS	EfficientNet / MobileNet / ResNet
IMAGE PROC.	OpenCV
EXPLAINABILITY	Grad-CAM
EDGE AI	TensorFlow Lite / ONNX Runtime
BACKEND	FastAPI (Python)
DATABASE	Firebase Auth + Firestore
STORAGE	Firebase Cloud Storage
DASHBOARD	React + Chart.js
HARDWARE	Smartphone + low-cost retinal attachment

CONNECTIVITY MODES	
OFFLINE	Capture → on-device AI → encrypted local storage (queued)
ONLINE	Local data → secure sync → cloud → doctor dashboard

IMPLEMENTATION METHODOLOGY	
Dataset & labels	Model training
Mobile app	Backend + cloud
Explainability + referral	Offline & field testing
MVP pilot	
Iterative build; each stage validated before the next. Clinician review remains in the loop at the referral stage.	

Technical Feasibility, Risks & Mitigation

NO ACCURACY OR CLINICAL PERFORMANCE CLAIMS – MODEL NOT YET TRAINED OR VALIDATED

AI MODEL PIPELINE – BUILT ON ESTABLISHED METHODS

Public retinal datasets > Preprocessing > Augmentation > Transfer learning > Validation > Deployment

CANDIDATE MODELS EfficientNet · MobileNet · ResNet METRICS Sensitivity & specificity first, then F1 / ROC-AUC

For a screening tool the cost of a miss outweighs the cost of a false alarm, so decision thresholds will be tuned toward sensitivity and every result stays subject to clinician review.

DEPLOYMENT PATHS

Cloud inference – for higher compute and larger models.

TFLite / ONNX on device – for supported Android hardware; this is what creates the offline-first pathway.

CHALLENGES & MITIGATION

Poor image quality

Risk – blurred or badly framed retinal images reach the model.

Mitigation – automated quality gate, guided capture overlay, retake prompt before inference.

Limited internet

Risk – rural connectivity interrupts cloud-based workflows.

Mitigation – offline-first local storage with edge inference and deferred, secure synchronisation.

Class imbalance

Risk – severe and proliferative classes have far fewer samples.

Mitigation – class weighting, balanced sampling, targeted augmentation, stratified splits.

False negatives

Risk – a missed high-risk case is clinically significant.

Mitigation – optimise screening sensitivity, threshold tuning, thorough validation, clinician review required for referral decisions.

Model generalisation

Risk – images from other devices and populations differ from training data.

Mitigation – diverse validation data, image normalisation, device and subgroup evaluation.

Privacy of patient data

Risk – retinal images and patient records are sensitive.

Mitigation – authentication, role-based access, encryption, data minimisation, consent and controlled access.

FEASIBILITY SCORECARD

TECHNICAL

Established CV/ML frameworks + transfer learning

HARDWARE

Commodity smartphone + low-cost optical attachment

SOFTWARE

Flutter + Python + FastAPI + Firebase

SCALABILITY

Cloud backend, modular services, multi-clinic ready

MVP

Capture + quality check + AI screening + referral

SECURITY & COMPLIANCE

Authentication

Role-based access

Encryption in transit & at rest

Data minimisation

Patient consent

Secure image storage

Audit logs

Clinician oversight

Responsible AI practices

ONE MISSION. BUILD THE IMPOSSIBLE.

Who Benefits, and How

**RURAL PATIENTS**
Access screening closer to home, without travelling to a specialist hospital first.

**HEALTH WORKERS**
A simple guided workflow – capture, quality check, result, referral – with no specialist training assumed.

**OPHTHALMOLOGISTS**
Remote review of images, heatmaps and risk bands, with referrals arriving already prioritised.

**PRIMARY HEALTH CENTRES**
Digital screening records and follow-up status for the population they serve.

IMPACT FLOW



Designed to support earlier detection and referral, potentially reducing delays in care. Screening outcomes remain subject to clinician evaluation.

SOCIAL BENEFITS

ACCESSIBILITY
Brings screening closer to underserved communities.

AFFORDABILITY
Uses commodity smartphone hardware rather than relying entirely on expensive dedicated imaging systems.

SPECIALIST REACH
Images and results can be reviewed remotely by specialists.

CONTINUITY
Digital screening history supports structured follow-up.

SCALABILITY
Additional clinics and health centres can connect to the same backend as the network grows.

ECONOMIC BENEFITS

- ▶ Potentially reduces unnecessary travel for preliminary screening.
- ▶ Makes better use of limited specialist capacity by prioritising who is seen first.
- ▶ Lowers the barrier to an initial screening contact.
- ▶ Potentially lowers per-site equipment cost compared with dedicated fundus cameras.

WHAT CHANGES ON THE GROUND



Environmental note – fewer avoidable trips for preliminary screening also means less travel per patient screened, a secondary environmental benefit rather than a primary design goal.

Research Basis

CLINICAL BACKGROUND & GRADING STANDARDS

WHO – Blindness and vision impairment (fact sheet). [who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment](https://www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment)

IAPB – Diabetic retinopathy resource hub. iapb.org/learn/resources/diabetic-retinopathy/

ICO – Guidelines for Diabetic Eye Care, International Council of Ophthalmology. [icoph.org](https://www.icoph.org)

Wilkinson CP et al. Proposed international clinical diabetic retinopathy and diabetic macular oedema severity scales. *Ophthalmology* 2003;110(9):1677-1682. – basis for the five screening categories

AI-BASED DR SCREENING

Gulshan V et al. Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs. *JAMA* 2016;316(22):2402-2410. [doi:10.1001/jama.2016.17216](https://doi.org/10.1001/jama.2016.17216)

Ting DSW et al. Deep learning system for diabetic retinopathy and related eye diseases using retinal images from multiethnic populations. *JAMA* 2017;318(22):2211-2223. [doi:10.1001/jama.2017.18152](https://doi.org/10.1001/jama.2017.18152)

Rajalakshmi R et al. Automated diabetic retinopathy detection in smartphone-based fundus photography using artificial intelligence. *Eye* 2018;32:1138-1144. [doi:10.1038/s41433-018-0064-9](https://doi.org/10.1038/s41433-018-0064-9) – precedent for smartphone-based capture

EXPLAINABILITY & MODEL ARCHITECTURES

Selvaraju RR et al. Grad-CAM: visual explanations from deep networks via gradient-based localisation. *ICCV* 2017. [arXiv:1610.02391](https://arxiv.org/abs/1610.02391)

Tan M, Le QV. EfficientNet: rethinking model scaling for convolutional neural networks. *ICML* 2019. [arXiv:1905.11946](https://arxiv.org/abs/1905.11946)

Sandler M et al. MobileNetV2: inverted residuals and linear bottlenecks. *CVPR* 2018. [arXiv:1801.04381](https://arxiv.org/abs/1801.04381)

DATASETS UNDER CONSIDERATION

EyePACS – Diabetic Retinopathy Detection, Kaggle. kaggle.com/competitions/diabetic-retinopathy-detection

APTOS 2019 – Blindness Detection, Kaggle. kaggle.com/competitions/aptos2019-blindness-detection

Datasets will be used only in accordance with their applicable licences and competition usage conditions; the team claims no ownership of them.

TECHNICAL DOCUMENTATION

PyTorch

TensorFlow / TF Lite

ONNX Runtime

OpenCV

FastAPI

Flutter

Firebase

React · Chart.js

REVIEWER NOTES

- This Round 1 submission presents the proposed solution, architecture and implementation plan; a live prototype is not required at this stage.

► RetinaSight AI is proposed as an AI-assisted screening and referral-support system, not a replacement for ophthalmologists.

► Public datasets will only be used in accordance with their applicable licences and usage conditions.
- No unverified model accuracy or clinical performance claims are made anywhere in this document.

► The MVP will prioritise retinal image capture, image-quality assessment, AI screening, explainability and referral support.

GENERATIVE AI DISCLOSURE

Generative AI tools were used as development and documentation assistants for brainstorming, structuring and refining portions of the project documentation and implementation planning. The final solution, technical decisions and submission were reviewed by the participating team.